ANGULARJS-III

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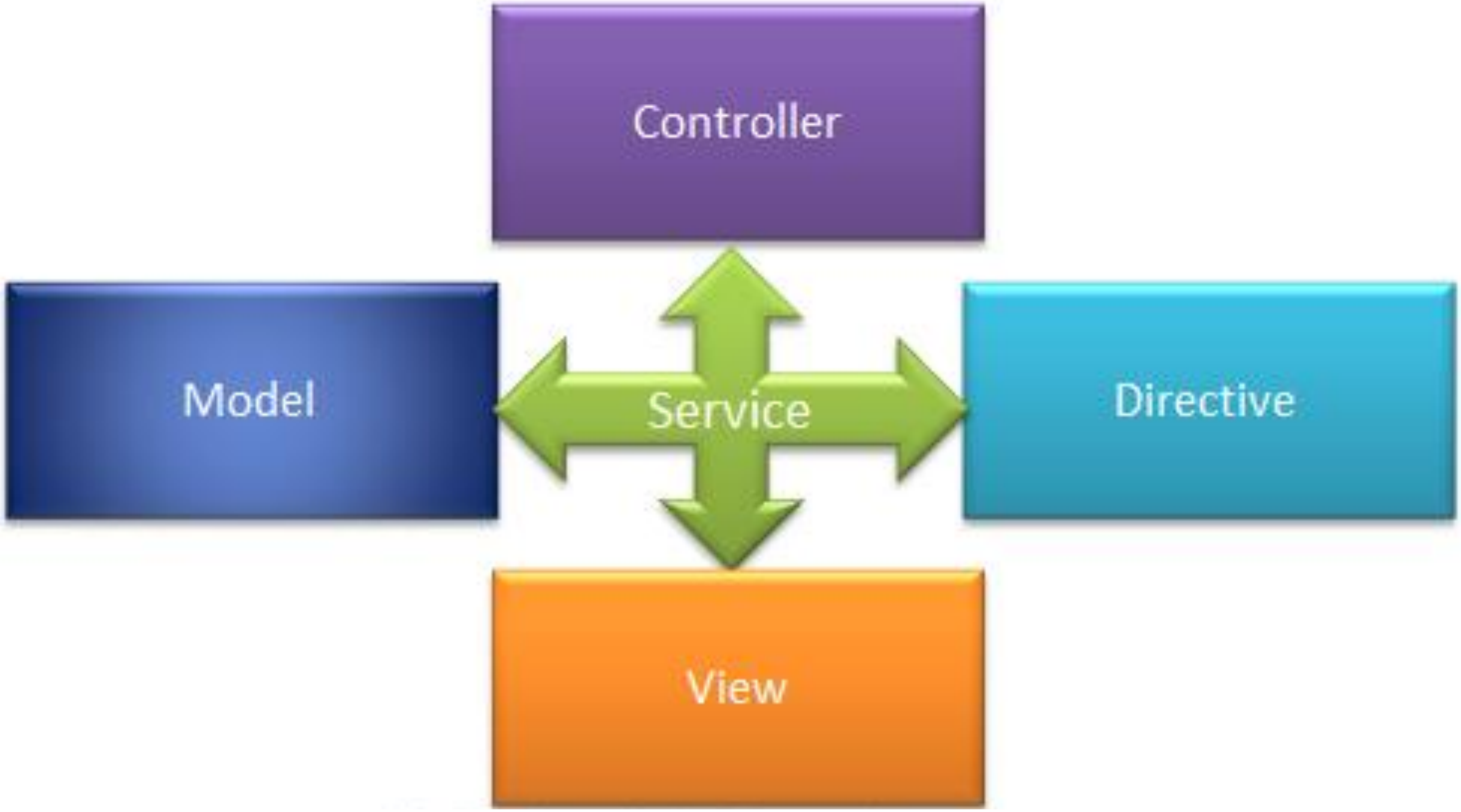
AngularJS Service

- AngularJS services are JavaScript functions for specific tasks, which can be reused throughout the application.
- AngularJS includes services for different purposes. For example, $http service can be used to send an AJAX request to the remote server.
- AngularJS also allows you to create custom service for your application.

AngularJS Service



Built-in AngularJS services

$anchorScroll	$exceptionHandler	$interval	$rootScope
$animate	$filter	$locale	$sceDelegate
$cacheFactory	$httpParamSerializer	$location	$sce
$templateCache	$httpParamSerializerJQLike	$log	$templateRequest
$compile	$http	$parse	$timeout
$controller	$httpBackend	$q	$window
$document	$interpolate	$rootElement	

$http Service

- The $http service is used to send or receive data from the remote server using browser's XMLHttpRequest or JSONP.

- $http is a service as an object.

$http Service

- It includes following shortcut methods.

Method	Description
$http.get()	Perform Http GET request.
$http.head()	Perform Http HEAD request.
$http.post()	Perform Http POST request.
$http.put()	Perform Http PUT request.
$http.delete()	Perform Http DELETE request.
$http.jsonp()	Perform Http JSONP request.
$http.patch()	Perform Http PATCH request.

$http.get()

```
<html>
<script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.6
.9/angular.min.js"></script>
<body>

<div ng-app="myApp" ng-controller="myCtrl">

<p>Today's welcome message is:</p>

<h1>{{myWelcome}}</h1>

</div>

<script>
var app = angular.module('myApp', []);
app.controller('myCtrl', function($scope, $http) {
  $http.get("welcome.htm")
  .then(function(response) {
      $scope.myWelcome = response.data;
  });
});
</script>

</body>
</html>
```

Output

Today's welcome message is:

Hello AngularJS Students

```
<!DOCTYPE html>
<html>
<script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.6.9/angula
r.min.js"></script>
<body>

<div ng-app="myApp" ng-controller="myCtrl">

<p>Today's welcome message is:</p>

<h1>{{myWelcome}}</h1>

</div>

<script>
var app = angular.module('myApp', []);
app.controller('myCtrl', function($scope, $http) {
  $http({
    method : "GET",
    url : "welcome.htm"
  }).then(function mySuccess(response) {
      $scope.myWelcome = response.data;
    }, function myError(response) {
      $scope.myWelcome = response.statusText;
  });
});
</script>

</body>
</html>
```

The example above executes the $http service with an object as an argument. The object is specifying the HTTP method, the url, what to do on success, and what to do on failure.

Output

Today's welcome message is:

Hello AngularJS Students

Properties

- The response from the server is an object with these properties:
  - .config the object used to generate the request.
  - .data a string, or an object, carrying the response from the server.
  - .headers a function to use to get header information.
  - .status a number defining the HTTP status.
  - .statusText a string defining the HTTP status.

```
<!DOCTYPE html>
<html>
<script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.6.9/angula
r.min.js"></script>
<body>

<div ng-app="myApp" ng-controller="myCtrl">

<p>Data : {{content}}</p>
<p>Status : {{statuscode}}</p>
<p>StatusText : {{statustext}}</p>

</div>



<script>
var app = angular.module('myApp', []);
app.controller('myCtrl', function($scope, $http) {
  $http.get("welcome.htm")
  .then(function(response) {
      $scope.content = response.data;
      $scope.statuscode = response.status;
      $scope.statustext = response.statusText;
  });
});
</script>

</body>
</html>
```

Output

Data : Hello AngularJS Students

Status : 200

StatusText :

To handle errors

- add one more function to the .then method:

```
var app = angular.module('myApp', []);
app.controller('myCtrl', function($scope, $http) {
  $http.get("wrongfilename.htm")
  .then(function(response) {
    // First function handles success
    $scope.content = response.data;
  }, function(response) {
    // Second function handles error
    $scope.content = "Something went wrong";
  });
});
```

Output:

Something went wrong

$location Service

- The $location service has methods which return information about the location of the current web page:

Use the `$location` service in a controller:

```
var app = angular.module('myApp', []);
app.controller('customersCtrl', function($scope, $location) {
    $scope.myUrl = $location.absUrl();
});
```

The $timeout Service

Display a new message after two seconds:

```
var app = angular.module('myApp', []);
app.controller('myCtrl', function($scope, $timeout) {
  $scope.myHeader = "Hello World!";
  $timeout(function () {
    $scope.myHeader = "How are you today?";
  }, 2000);
});
```

The $interval Service

Display the time every second:

```
var app = angular.module('myApp', []);
app.controller('myCtrl', function($scope, $interval) {
  $scope.theTime = new Date().toLocaleTimeString();
  $interval(function () {
    $scope.theTime = new Date().toLocaleTimeString();
  }, 1000);
});
```

Guess The Output Of the Program

```
<!DOCTYPE html>
<html >
<head>
    <script src="~/Scripts/angular.js"></script>
</head>
<body ng-app="myApp">
    <div ng-controller="myController">
        {{counter}}
    </div>
    <script>
        var myApp = angular.module('myApp', []);

        myApp.controller("myController", function ($scope, $interval) {
            $scope.counter = 0;

            var increaseCounter = function () {
                $scope.counter = $scope.counter + 1;
            }

            $interval(increaseCounter, 1000);
        });
    </script>
</body>
</html>
```

$window Service

- AngularJs includes $window service which refers to the browser window object.

- In the JavaScripts, window is a global object which includes many built-in methods like alert(), prompt() etc.

- It is recommended to use $window service in AngularJS

```html
<!DOCTYPE html>
<html>
<head>
    <script src="~/Scripts/angular.js"></script>
</head>
<body ng-app="myApp" ng-controller="myController">
    <button ng-click="DisplayAlert('Hello World!')">Display Alert</button>
    <button ng-click="DisplayPrompt()">Display Prompt</button>
    <script>
        var myApp = angular.module('myApp', []);

        myApp.controller("myController", function ($scope, $window) {

            $scope.DisplayAlert = function (message) {
                $window.alert(message);
            }

            $scope.DisplayPrompt = function () {
                var name = $window.prompt('Enter Your Name');

                $window.alert('Hello ' + name);
            }

        });
    </script>
</body>
</html>
```

AngularJS Filters

- AngularJS Filters allow us to format the data to display on UI without changing original format.

- Filters can be used with an expression or directives using pipe | sign.

- {{expression | filterName:parameter }}

Important filters

Filter Name	Description
Number	Formats a numeric data as text with comma and fraction.
Currency	Formats numeric data into specified currency format and fraction.
Date	Formats date to string in specified format.
Uppercase	Converts string to upper case.
Lowercase	Converts string to lower case.
Filter	Filters an array based on specified criteria and returns new array.
orderBy	Sorts an array based on specified predicate expression.
Json	Converts JavaScript object into JSON string
limitTo	Returns new array containing specified number of elements from an existing array.

Number Filter

- A number filter formats numeric data as text with comma and specified fraction size.
- {{ number_expression | number:fractionSize}}
- If a specified expression does not return a valid number then number filter displays an empty string.

Number Filter

Example: Number filter

```
<!DOCTYPE html>
<html >
<head>
    <script src="~/Scripts/angular.js"></script>
</head>
<body ng-app >
    Enter Amount: <input type="number" ng-model="amount" /> <br />

    100000 | number = {{100000 | number}} <br />
    amount | number = {{amount | number}} <br />
    amount | number:2 = {{amount | number:2}} <br />
    amount | number:4 = {{amount | number:4}} <br />
    amount | number = <span ng-bind="amount | number"></span>
</body>
</html>
```

Result:

AngularJS Number Filter

Enter Amount: 786

100000 | number = 100,000

amount | number = 786

amount | number:2 = 786.00

amount | number:4 = 786.0000

amount | number = 786

Currency Filter

- The currency filter formats a number value as a currency.
- When no currency symbol is provided, default symbol for current locale is used.
- {{ expression | currency : 'currency_symbol' : 'fraction'}}

Example: Currency filter

```
<!DOCTYPE html>
<html >
<head>
    <script src="~/Scripts/angular.js"></script>
</head>
<body ng-app="myApp">
    <div ng-controller="myController">
        Default currency: {{person.salary | currency}} <br />
        Custom currency identifier: {{person.salary | currency:'Rs.'}} <br />
        No Fraction: {{person.salary | currency:'Rs.':0}} <br />
        Fraction 2: <span ng-bind="person.salary| currency:'GBP':2"></span>
    </div>
    <script>
        var myApp = angular.module('myApp', []);

        myApp.controller("myController", function ($scope) {
            $scope.person = { firstName: 'James', lastName: 'Bond', salary: 100000}
        });
    </script>
</body>
</html>
```

Result:

```
Default currency: $100,000.00
Custom currency identifier: Rs.100,000.00
No Fraction: Rs.100,000
Fraction 2: GBP100,000.00
```

Date filter

- Formats date to string based on the specified format.

- {{ date_expression | date : 'format'}}

Example: date filter

```
<!DOCTYPE html>
<html >
<head>
    <script src="~/Scripts/angular.js"></script>
</head>
<body ng-app>
    <div ng-init="person.DOB = 323234234898">
        Default date: {{person.DOB| date}} <br />
        Short date: {{person.DOB| date:'short'}} <br />
        Long date: {{person.DOB | date:'longDate'}} <br />
        Year: {{person.DOB | date:'yyyy'}} <br />
    </div>
</body>
</html>
```

Result:

```
Default date: Mar 30, 1980
short date: 3/30/80 8:47 AM
long date: March 30, 1980
Year: 1980
```

Uppercase/lowercase filter

Example: uppercase & lowercase filters

```
<!DOCTYPE html>
<html >
<head>
    <script src="~/Scripts/angular.js"></script>
</head>
<body ng-app>
    <div ng-init="person.firstName='James';person.lastName='Bond'">
        Lower case: {{person.firstName + ' ' + person.lastName | lowercase}} <br />
        Upper case: {{person.firstName + ' ' + person.lastName | uppercase}}
    </div>
</body>
</html>
```

Result:

```
Lower case: james bond
Upper case: JAMES BOND
```

Filter

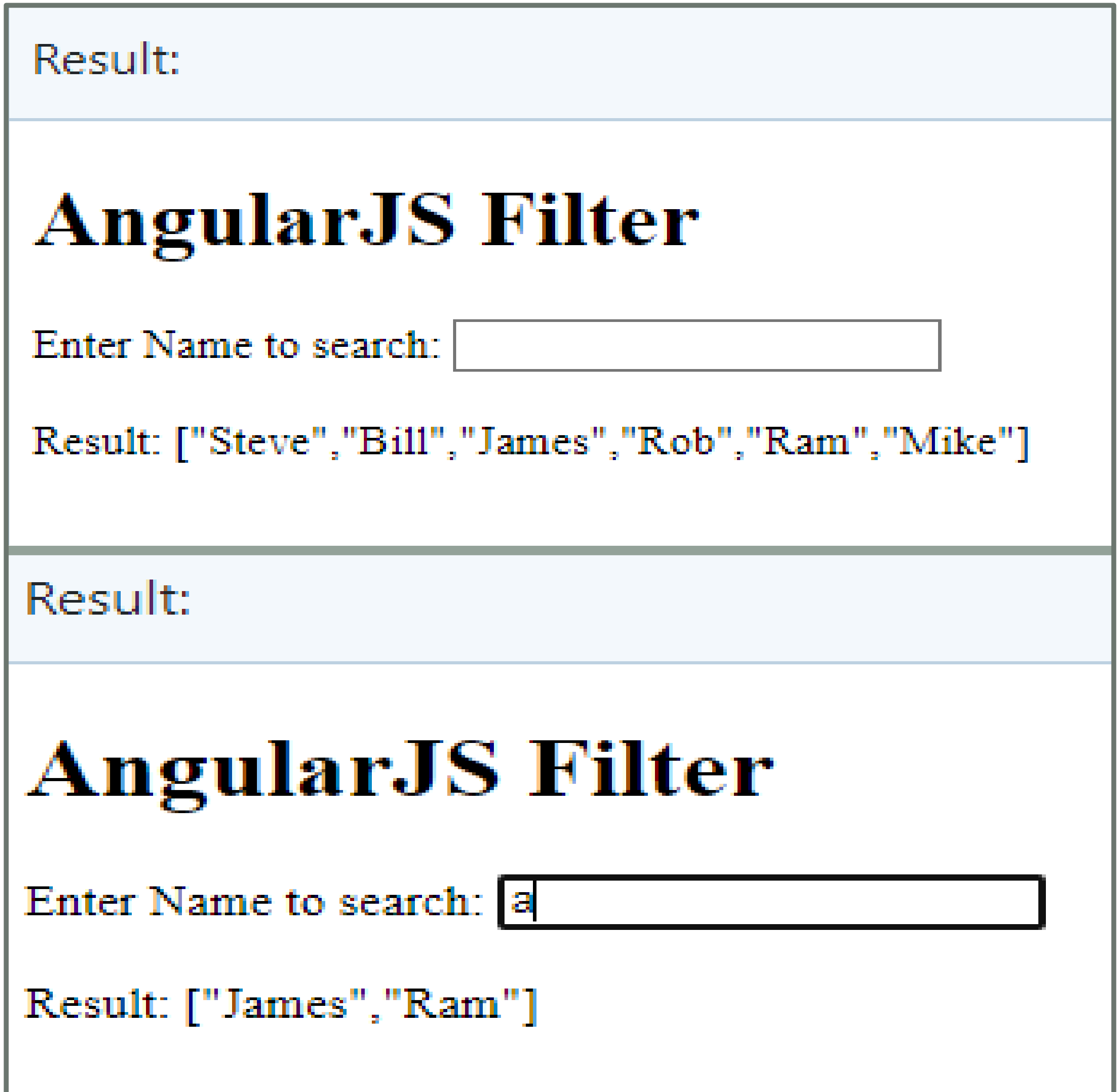
- Filter selects items from an array based on the specified criteria and returns a new array.

- {{ expression | filter : filter_criteria }}

Example: filter

```
<!DOCTYPE html>
<html >
<head>
    <script src="~/Scripts/angular.js"></script>
</head>
<body ng-app="myApp">
    <div ng-controller="myController">
        Enter Name to search: <input type="text" ng-model="searchCriteria" /> <br />
        Result: {{myArr | filter:searchCriteria}}
    </div>
    <script>
        var myApp = angular.module('myApp', []);

        myApp.controller("myController", function ($scope) {
            $scope.myArr = ['Steve', 'Bill', 'James', 'Rob', 'Ram', 'Moin']
        });
    </script>
</body>
</html>
```


Result:
AngularJS Filter
Enter Name to search:
Result: ["Steve","Bill","James","Rob","Ram","Mike"]
Result:
AngularJS Filter
Enter Name to search: a
Result: ["James","Ram"]

Thank you students